# Nikhil Gavel

Computer Science Undergraduate | Software Engineering Intern

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LinkedIn | GitHub | Portfolio | Codeforces

## Education

**International Institute of Information Technology, Naya Raipur** — Naya Raipur, India

*B.Tech., Computer Science and Engineering* — Aug 2026 – May 2030

Relevant coursework: Programming in C, Data Structures and Algorithms, Calculus, Linear Algebra, Digital Electronics (Verilog), Internet of Things.

## Technical Skills

**Languages:** C, Python, SQL; working knowledge of Rust, C++, JavaScript and Bash
**Frameworks and tools:** Tauri v2, React, FastAPI, Git, GitHub, Docker Compose, Linux/Unix, REST APIs
**Data and testing:** SQLite, PostgreSQL, manual and exploratory testing, functional testing, defect reporting, `pytest`
**Development workflow:** AI-assisted implementation, code review and diff auditing, debugging, performance profiling

## Projects

**WatchMark – Desktop Media Tracker** — 2026

*Rust, Tauri v2, React, TypeScript, SQLite / GitHub*

- Built a Windows desktop companion for VLC that tracks playback progress, organizes local media, and stores viewing history in a local SQLite database.
- Worked across a modular Rust/Tauri backend and React/TypeScript interface, including VLC local-HTTP integration, TMDb metadata lookup, filesystem scanning, and SQLite persistence.
- Used AI coding tools as implementation assistance; reviewed generated diffs, checked feature completeness, and tested builds on Windows hardware.
- Investigated a WebView2 GPU-utilization issue associated with animated UI effects; changed the animation approach and measured idle GPU use fall from 70.3% to approximately 0–1% in the tested setup.
- Measured idle memory use at approximately 15.1 MB and idle CPU use at approximately 0% on the test machine; results depend on hardware and runtime conditions.

## Experience

**test IO (EPAM Systems)** — Remote

*Freelance QA Tester* — Jan 2026 – Present

- Perform functional and exploratory testing of web and mobile products, checking user flows, responsive layouts, and behavior on physical devices.
- Submitted 13 defects across three test cycles; 11 were approved (85% approval rate, based on current personal tracking).
- Write reproducible defect reports with steps, expected and actual results, and supporting details; reported issues included navigation failures, mobile layout defects, and validation problems.
- Received a five-star test-cycle rating (10/10 acceptance, as recorded in the test-cycle feedback).

## Achievements

- **Indian Navy THINQ 2024:** National semifinalist (Top 16), representing Sri Sankara Vidyalaya; competition materials cited 12,600+ participating school teams.
- **Competitive programming:** Solved 130+ problems across GeeksforGeeks, LeetCode, and Codeforces.
- **School academics:** Class XII – 91.6%; Class X – 94.0%.

## Languages

English (Professional working proficiency) | Hindi (Native)